MIL-B-16028F

9 May 1974
SUPERSEDING
MIL-B-16028E
6 October 1970

MILITARY SPECIFICATION

BUCKETS, CLAMSHELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers clamshell buckets.

1.2 Classification. Bucket shall be one of the following types and sizes, as specified (see 6.2.1):

Type I - Light duty rehandling.
Type II - General purpose.
Type III - Heavy-duty excavation.
Type IV - Ore rehandling.

Size 38 - 3/8 cubic yard (cu yd).
Size 50 - 1/2 cu yd.
Size 75 - 3/4 cu yd.
Size 100 - 1 cu yd.
Size 150 - 1-1/2 cu yd.
Size 200 - 2 cu yd.
Size 250 - 2-1/2 cu yd.
Size 350 - 3-1/2 cu yd.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.
MIL-C-3580 - Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full-Revolving, and Their Attachments, Packaging of.

FSC 3815

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MIL-T-52176 - Teeth, Surface Ripping; (Points, Shanks, Keepers, and Keeper Locks), Bucket and Dipper, Crane-Shovel.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standard required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Description. The bucket shall be of the two-line clamshell type, operated by the multiple line block and tackles, or by lever arm type arrangement. The bucket unit shall be complete and ready for use.

* 3.2 First article. When specified (see 6.2.1), the contractor shall furnish a bucket for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard bucket, components, and accessories. Except as specified in 3.6.1 through 3.6.10, the bucket, components, and accessories shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification.

* 3.4 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly.

3.5 Capacities and weights.

3.5.1 Plate line capacity. The plate line capacity shall be defined as the number of cubic feet (cu ft) the bucket will hold when heaped with a fine granular material, and the surplus stricken off by drawing a straightedge over the top of the scoop. Plate line capacities shall be not less than that specified in table I.

Table I. Plate line capacities

	Size							
	38	50	75	100	150	200	250	350
Plate line capacity (cu ft)	10.0	13.5	19.0	26.5	39.0	52.0	64.0	85.5

3.5.2 Weights of buckets and counterweights. The weight of the bucket without teeth shall be not less than that specified in table II. The weight of the counterweights shall not exceed the given percentage of the total weight of bucket and counterweights shown in table II for the various sizes of type II, III, and IV buckets.

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Table II. Bucket weight

	Size							
	38	50	75	100	150	200	250	350
Type I, weight of bucket only, min (lb)	1,100	1,500	2,300	2,800	4,000	5,100	6,200	7,800
Type II, weight of bucket and counterweights, min (lb)	1,500	2,500	3,000	3,700	5,550	6,650	7,950	9,550
Type II, max percent counterweights of total weight (percent)	21	17	14	13	12	10	10	10
Type III, weight of bucket and counterweights, min (lb)	1,900	2,750	3,800	4,450	6,150	7,650	9,150	10,450
Type III, max percent counterweights of total weight (percent)	18	14	12	12	12	12	11	9
Type IV, weight of bucket and counterweights, min (lb)				6,000	8,600	10,200	12,700	15,000
Type IV, max percent counterweights of total weight (percent)				13	12	12	10	8

3.6 Components.

3.6.1 Bucket shell. The bucket shell shall consist of two scoops hinged together. Tagline attachments shall be furnished. The scoops shall be designed in accordance with best commercial practice to handle efficiently the material for which the bucket is intended. The inside surfaces shall be smooth and free from rivet or bolt heads and other projections. The scoop at the corner brackets shall be of sufficient cross

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section and strength to permit full transference of maximum design stress to the corner arms. Scoop plate material shall be steel which meets or exceeds the requirements of ASTM Designation A36. Scoop plates shall be joined by either riveting or welding unless of the one-piece type.

* 3.6.2 Lips. The lips shall form the cutting edge of the scoops, and shall extend in one piece across the bottom and up both sides to the top of the scoop. The lips shall be of adequate width and thickness to provide for secure attachment of the teeth. Lips shall be of abrasion-resistant alloy steel of the type that has been proven satisfactory in extensive use, and shall be heat treated to not less than 185 Brinell hardness. The design of all lips or cutting edges shall be such as to assure proper fit, eliminate resistance for material entering the scoop, and permit replacement by welding or riveting.

3.6.3 Teeth. Unless otherwise specified (see 6.2.1), manufacturer's standard bucket teeth shall be furnished. When specified (see 6.2.1), teeth, including points, shanks, keepers, and keeper locks, shall conform to MIL-T-52176, as applicable. Type II, III, and IV buckets, and when specified (see 6.2.1), type I bucket shall be provided with not less than five bottom teeth. When specified (see 6.2.1), manufacturer's standard side and corner teeth shall be furnished. The teeth shall be readily replaceable by the use of standard tools. The shanks shall be attached to the bucket with plow bolts fabricated from steel having a minimum tensile strength of not less than 105,000 pounds per square inch.

3.6.4 Hinge. The hinge shall be of steel, either of the single or double shaft type, and arranged for multiple line or lever arm type closing action. In the multiple line block and tackle type of closing action, the upper block shall be attached to the bucket head and the lower block attached to the hinge body or the lever arm. In the lever arm type of closing action, the lever arm or arms shall be either integral with or rigidly secured to the hinge body or to one or both of the scoops. Where the single shaft design is used, the scoops that rotate on the shaft shall be equipped with replaceable bushings. Where the double shaft design is used, both scoops shall be pinned to the shafts which, in turn, shall rotate on replaceable bushings. Provision shall be made for attaching counterweights in accordance with good commercial practice. The reeving of the closing line shall avoid reverse bends over the sheaves. Stops shall be provided for the correct digging position of the scoops.

3.6.5 Bucket head. The bucket head shall be a steel casting or weldment with attachments for the corner arms, closing line sheaves, and sockets for the hoisting and, when necessary, the closing line.

3.6.6 Corner arms. Corner arms shall be one-piece forged steel, or of bar steel with welded or upset bosses.

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3.6.7 Brackets. The hinge and arm brackets shall be of steel and of sufficient size and strength to accommodate stresses caused by lifting, dropping, bending, or twisting forces.

* 3.6.8 Sheaves. Unless otherwise specified (see 6.2.1), sheaves shall be provided with manufacturer's current standard bearings or bushings. Sheaves shall have smooth grooves and a pitch diameter of not less than 15 times the diameter of the largest wire rope recommended by the bucket manufacturer for use with the bucket. When specified (see 6.2.1), sheave groove cleaners shall be furnished.

3.6.9 Rollers, shafts, and pins. Rollers, shafts, and pins shall be the manufacturer's current standard.

3.6.10 Bushings. Bushings shall be heat-treated alloy steel, alloy bronze, or cast iron, as applicable.

3.7 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2.1), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2.1), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. In either case, color of the finish coat shall be as specified (see 6.2.1).

* 3.7.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, mill scale which can be removed by wire brushing, products of corrosion, dirt, or other foreign substances.

3.7.2 Treatment. As soon as practicable after cleaning, and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system.

3.7.3 Painting. Painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, orange peel, or other defects.

3.8 Nameplate. A nameplate shall be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose and securely affix it to each unit in a conspicuous place with rivets, nonferrous screws, or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

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3.9 Lubrication. Lubrication fittings shall be provided in accessible, protected positions for bearings and moving parts requiring lubrication.

3.10 Workmanship. All parts, components, and assemblies of the bucket, including castings, forgings, molded parts, stamping, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

* 3.10.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.10.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

* 3.10.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

* 3.10.4 Welders and welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Unless otherwise specified (see 6.2.1), welding procedures shall be in accordance with a nationally recognized welding code. When specified (see 6.2.1), before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by either the American Welding Society or the American Society of Mechanical Engineers for the type of welding operation to be performed.

* 3.10.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards.

* 4.2 Examination. Each bucket shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.2.1 First article inspection. First article inspection shall be performed on one bucket when a first article sample is required (see 3.2). This inspection shall include the examination of 4.2 and the test of 4.3. The first article may be a standard production item from the supplier's current inventory provided the bucket meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining buckets to be furnished under the contract.

4.3 First article tests. When first article is required, the first article bucket of each type and size shall be subjected to the tests specified in 4.3.1 through 4.3.3 as applicable. The first article tests shall be performed by the supplier. The supplier shall provide all material and equipment to perform all testing specified. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all buckets until corrective action has been taken.

* 4.3.1 Materials rehandling tests. Type I bucket procured without teeth shall be subjected to materials rehandling tests. Material for

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testing shall be loose or stockpiled earth, sand, gravel, or other comparable material. The open bucket shall be set down on the material, closed without interruption, hoisted, swung to the side, and dumped. The average load from 10 cycles shall be at least equal to plate line capacity of the bucket.

4.3.2 Digging tests.

* 4.3.2.1 Type III bucket. Type III bucket shall be subjected to digging tests. Material for testing shall be in-place dry clay or soft shale. The surface of the test material shall be relatively smooth with not over 2 percent average grade. The open bucket shall be set down on the material. Power shall be applied to the closing line until the bucket is closed. Closing shall be positive and uninterrupted. The bucket shall be raised and conveyed to the proper location where material shall be completely dumped. The average load from 10 cycles shall be at least equal to plate line capacity of the bucket.

* 4.3.2.2 Type IV bucket. Type IV bucket shall be subjected to tests specified in 4.3.2.1 except that the test material shall be loose or stockpiled earth, rock, or a combination.

* 4.3.3 Combination tests. Type I bucket procured with teeth and type II bucket shall be subjected to the following tests:

(a) With teeth installed, the bucket shall be tested as specified in 4.3.2.1, except that the test material shall be moist clay or moist loam.
(b) With teeth removed, the bucket shall be tested as specified in 4.3.1.

4.4 Production unit tests. The contractor's testing system shall, as a minimum, assure that the bucket is capable of meeting the requirements of this specification.

4.5 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the buckets shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A, B, or C, as specified (see 6.2.1).

5.1.1 Level A. The complete buckets shall be preserved and packaged in accordance with level A of MIL-C-3580.

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5.1.2 Level B. The complete buckets shall be preserved and packaged in accordance with level B of MIL-C-3580.

5.1.3 Level C. The complete buckets shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. The complete buckets shall be packed in accordance with level A of MIL-C-3580.

5.2.2 Level B. The complete buckets shall be packed in accordance with level B of MIL-C-3580.

5.2.3 Level C. The buckets shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages, shipping containers, and the buckets shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers the standard types of clamshell buckets to be used for earth excavation and handling of loose materials.

6.1.1 Type I, light duty rehandling. This type is intended for handling all kinds of loose materials, including concrete aggregate. Limiting size for crushed rock or gravel is approximately 2-1/2 inches.

6.1.2 Type II, general purpose. This type is intended for excavating moist clay, loam, sand, and loose gravel. By removing the teeth, the bucket may be used for rehandling materials.

6.1.3 Type III, heavy-duty excavation. This type is intended for excavating dry clay, gravel, or soft shale.

6.1.4 Type IV, ore rehandling. This type is intended for rehandling and stockpiling ore.

6.2 Ordering data. Procurement documents should specify:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Type and size of bucket required (see 1.2).

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(c) When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
(d) When teeth, points, shanks, keepers, and keeper locks shall conform to MIL-T-52176 (see 3.6.3).
(e) When type I bucket with teeth is required (see 3.6.3).
(f) When side and corner teeth are required (see 3.6.3).
(g) When sheave bearings other than specified are required (see 3.6.8).
(h) When sheave groove cleaners are required (see 3.6.8).
(i) When cleaning, treatment, and painting in accordance with MIL-T-704, type A is required (see 3.7).
(j) Finish coat color required (see 3.7).
(k) When certification of welder's qualification is required (see 3.10.4).
(l) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

* 6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data requirement	Applicable DD 1664
3.10.4	Certificate of Compliance	UDI-E-24007

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item consisting of one complete bucket unit or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications,

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corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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